

A. Quiz 2

Graph Theory. All graphs are assumed to be simple.

1. (1 marks) If a graph G is isomorphic to its own complement G^c and $|V(G)| = 21$, what is the value of $|E(G)|$?

Answer: $|E(G)| = 105$.

Explanation: Since $G \cong G^c$, we have $|E(G)| = |E(G^c)|$. Also $|E(G)| + |E(G^c)| = \binom{21}{2}$. Hence $2|E(G)| = \binom{21}{2} \implies |E(G)| = 105$.

2. (1 marks) Consider the following statement: “In a connected graph with n vertices and $n - 1 + k$ edges, the total number of distinct cycles is k .” For which values of k is this always true?

- (a) 0
- (b) 1
- (c) 2
- (d) all values of k

Answer: (a),(b)

Explanation: $k = 0$ corresponds to tree which has $n - 1$ edge. Let G be any connected graph with n vertices and n edges. Let T be a spanning tree of G which must have $n - 1$ edges. So G can be obtained from T by adding one edge. It is a property of tree that adding an edge creates exactly one cycle. For $k \geq 2$, number of cycles is at least k not exactly k .

3. (1 marks) A saturated hydrocarbon is a molecule C_mH_n in which every carbon atom has bonds with exactly 4 other atoms, every hydrogen atom has bond with exactly 1 other atom, and no subset of bonds form a cycle. Which of the following molecular formulas cannot represent a saturated hydrocarbon?

- (a) $C_{20}H_{42}$
- (b) $C_{21}H_{42}$
- (c) $C_{21}H_{20}$
- (d) $C_{22}H_{42}$

Answer: b, c, d

Explanation: The molecular graph must be connected, so a saturated hydrocarbon corresponds to a tree. The number of edges equals half the sum of vertex degrees, that is,

$$|E| = \frac{1}{2}(4m + n),$$

and for a tree we also have $|E| = m + n - 1$. Equating these gives $m = 2n + 2$. Conversely, for every n , a tree can be easily constructed corresponding to the molecule C_nH_{2n+2} , so this condition is both necessary and sufficient for a saturated hydrocarbon.

4. (1.5 marks) Let G be an Eulerian graph (i.e., G has an Eulerian tour). Recall the notation that for any edge $e \in E(G)$, $G \setminus \{e\}$ is a graph obtained by removing an edge e from G . Which of the following statements are always correct?
- (a) $G \setminus \{e\}$ is connected for all $e \in E(G)$.
 - (b) There exists an edge $e \in E(G)$ such that $G \setminus \{e\}$ is connected, but $G \setminus \{e\}$ is not connected for all $e \in E(G)$.
 - (c) The sum of degrees of all vertices in $G \setminus \{e\}$ is odd.
 - (d) The edge set $E(G)$ can be partitioned into edge-disjoint cycles (two cycles may share a vertex but not an edge).

Answer: (a), (d)

Explanation: In an Eulerian graph, every vertex has even degree, so option (c) is wrong because removing one edge decreases the sum of degrees by 2. Recall the proof that a connected graph is Eulerian if and only if every vertex has even degree: we first show that such a graph contains a cycle, then remove the edges of this cycle and apply the argument recursively to the remaining subgraph. The cycles obtained in each iteration are edge-disjoint (since we delete their edges before the next step), and together they form a partition of the edge set because the process continues until no edges remain. Hence option (d) is also correct. Finally, since every edge lies on a cycle, $G \setminus \{e\}$ is connected for all $e \in E(G)$. So option (a) is correct, but not (b).

5. (1 marks) Let M_1 and M_2 be two distinct matchings in a graph G . Consider a graph $D = M_1 \triangle M_2$, obtained by removing all edges of G that are either present in both M_1 and M_2 or not present in both M_1 and M_2 . Which of the following is **not** a possible degree of any vertex in D ?
- (a) 6
 - (b) 4
 - (c) 5
 - (d) 1

- (e) 2
- (f) 3

Answer: a, b, c, f

Explanation: Degrees 0, 1, 2 are possible. We have seen this in the proof of ‘A matching is maximum iff there is no augmenting path’.

6. (1.5 marks) In the above question, assume M_1 and M_2 are two distinct perfect matchings in G . Which of the following is **not** a possible degree of any vertex in D ?

- (a) 6
- (b) 4
- (c) 5
- (d) 1
- (e) 2
- (f) 3

Answer: a, b, c, d, f .

Explanation: In this case, even a vertex of degree 1 is not possible. Every vertex v is covered in both perfect matchings. There are two possibilities:
 (i) the edges incident to v in the two perfect matchings are the same, or
 (ii) the edges are different.

In the first case, $\deg_D(v) = 0$. In the second case, two distinct edges incident on v (one from each matching) remains, hence $\deg_D(v) = 2$. Therefore, only degrees 0 and 2 are possible.

7. (1.5 marks) The line graph $L(G)$ of a graph G is defined as follows: every vertex of $L(G)$ corresponds to an edge of G , and two vertices of $L(G)$ (which are edges in G) are adjacent if and only if the corresponding edges in G share a vertex. Which of the following statements are always true?

- (a) If G is Eulerian then $L(G)$ is Hamiltonian.
- (b) If $L(G)$ is Hamiltonian then G is Hamiltonian.
- (c) If G is Eulerian then $L(G)$ is Eulerian.

Answer: (a),(c)

Explanation: If G is Eulerian, each vertex has even degree. Then in $L(G)$, degree of vertex (edge uv of G) is $\deg(u) + \deg(v) - 2$, which is even, so $L(G)$ is Eulerian. The Eulerian tour of G gives a Hamiltonian cycle in $L(G)$. So (a),(c) are correct.

(b) can not be correct because we just showed that even for Eulerian graph (which may not be Hamiltonian), $L(G)$ is Hamiltonian.

8. (1.5 marks) Two players alternately select *distinct* vertices $v_0, v_1, v_2, \dots$ in a graph G such that for all $i > 0$, v_i is adjacent to v_{i-1} . Player 1 chooses v_0 . The game ends when a player is unable to select a new vertex. The last player who successfully selects a vertex wins. Give a necessary and sufficient condition on G under which Player 1 has a winning strategy.

Answer: Player 1 has a winning strategy if and only if G does not have a perfect matching.

Explanation: Suppose G has a perfect matching. Player 2 adopts the following strategy: whenever Player 1 selects a vertex v , Player 2 responds by choosing the vertex that is matched with v in the perfect matching. This ensures that Player 2 always has a legal move and will make the last move, thus winning the game.

Now suppose G does not have a perfect matching. We will show that Player 1 has a winning strategy. Let M be a maximum matching in G , and let V and V' denote the sets of matched and unmatched vertices under M , respectively. Since M is not perfect, V' is nonempty. Note that no two vertices in V' can not be adjacent, because such an edge would allow us to enlarge M , contradicting its maximality.

Player 1 starts the game by selecting a vertex in V' . Then Player 2 either immediately loses (if there is no adjacent vertex to move to), or is forced to choose a vertex in V . After that, Player 1 simply chooses the vertex that is matched with Player 2's last move in M . By continuing this strategy, Player 2 will never get a chance to choose a vertex from V' ; otherwise, there would exist an augmenting path, contradicting the maximality of M . Hence Player 1 will make the last move and win the game.

9. (1.5 marks) Let G be a bipartite graph with 50 vertices and 9 connected components. Degree of every vertex in G is 2. How many perfect matching does G have?

Answer: $2^9 = 512$.

Explanation: Since degree of every vertex is 2, this graph will be disjoint union of cycles. Further, as the graph is bipartite, each cycle will be of even length. In each component graph, 2 perfect matchings are possible. Therefore, total number of perfect matchings is $= 2^9$.

10. (1.5 marks) A truncated $n \times n$ chessboard is obtained by removing some squares from an $n \times n$ chessboard. Give the description of a bipartite graph in which the existence of a perfect matching will imply that the truncated chessboard can be tiled using 1×2 tile.

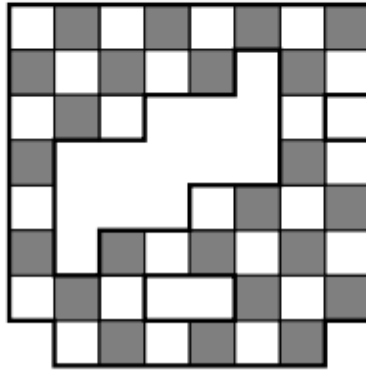


Figure 1: An example of a Truncated chessboard

Answer: Consider a bipartite graph $(X \cup Y, E)$ where X is all white squares and Y is all black squares present in the truncated board. If a white square is adjacent (share a side) to a black square in the truncated board, we add an edge between these two squares.

Explanation: We can place 1×2 tile on every adjacent white–black pair whose corresponding edge in the bipartite graph belongs to the perfect matching. By the definition of perfect matching, it is easy to see that no square will be left covered, neither two 1×2 tiles will get placed on one another.

11. (1.5 marks) Consider a bipartite graph $G = (X \cup Y, E)$ with $|X| = 10$ and $|Y| = 10$. Suppose that for every subset $S \subseteq X$, we have

$$|N(S)| \geq |S| - 2.$$

Further, there exists a subset S such that $N(S) = |S| - 2$. Which of the following statements is/are true?

- (a) A matching of size 8 is not guaranteed.
- (b) A matching of size 8 is guaranteed, and depending on the graph, there can or can not be a matching of size 9.
- (c) A matching of size 8 is guaranteed and a matching of size 9 is not possible.
- (d) A matching of size 7 is guaranteed, and depending on the graph, there can or can not be a matching of size 8.

Answer: (c)

Explanation: Let $|X| = |Y|$ and define

$$d = \max_{S \subseteq X} (|S| - |N(S)|).$$

We will show that there exists a matching of size $|X| - d$, and that no matching of size $|X| - d + 1$ can exist.

Add d new vertices on the Y side, and connect each of them to every vertex in X . For every subset $S \subseteq X$, the neighborhood size $|N(S)|$ increases by d , since all the new vertices are adjacent to every vertex in X . Hence, in the new graph, we have

$$|N(S)|_{\text{new}} = |N(S)| + d \geq |S|,$$

so by Hall's theorem, the new graph has an X -perfect matching. Each of the d new vertices can be incident to at most one edge of this perfect matching, so the number of matched edges that use only original vertices is at least $|X| - d$. Therefore, the original graph G has a matching of size at least $|X| - d$.

Now for the other part, our assumption is that there exists a subset S' such that $|N(S')| = |S'| - d$. So at most $|S'| - d$ vertices in S' can be covered by any matching. In other words, at least d vertices in S will not be matched by any matching. Thus maximum matching size is at most $|X| - d$.

Number Theory

1. (1 marks) Find the inverse of 95 modulo 111.

Answer: 104. (Marks will be provided as long as you have written any number in $\{\dots, -7, 104, 104 + 111, \dots\}$)

Explanation: We solve for integers x, y with

$$95x + 111y = 1.$$

Perform the Euclidean algorithm:

$$111 = 95 \cdot 1 + 16,$$

$$95 = 16 \cdot 5 + 15,$$

$$16 = 15 \cdot 1 + 1,$$

$$15 = 1 \cdot 15 + 0.$$

Back-substitute to express 1 as a combination of 95 and 111:

$$1 = 16 - 15,$$

$$= 16 - (95 - 16 \cdot 5) = 16 \cdot 6 - 95,$$

$$= (111 - 95) \cdot 6 - 95 = 111 \cdot 6 - 95 \cdot 7.$$

Thus

$$-95 \cdot 7 + 111 \cdot 6 = 1 \implies 95 \cdot (-7) \equiv 1 \pmod{111}.$$

2. (1 marks) Find the remainder when 7^{479} is divided by 29.

Answer: 24.

Explanation: By Fermat's Little Theorem, for prime 29 we have $7^{28} \equiv 1 \pmod{29}$. Write

$$479 = 28 \cdot 17 + 3.$$

Thus

$$7^{479} \equiv (7^{28})^{17} \cdot 7^3 \equiv 1^{17} \cdot 7^3 \equiv 7^3 \pmod{29}.$$

By direct computation, we get $7^3 = 24 \pmod{29}$

3. (1.5 marks) A band of 13 pirates steals some number of gold coins. When they try to divide the coins equally, 8 coins remain. A fight begins and one pirate is killed; the remaining pirates divide the coins and 5 coins remain. Another fight begins and again one pirate is killed; the remaining 11 pirates find that the coins divide evenly. Assume the total number of coins is at most 1716. Determine the number of stolen coins.

Answer: 1529